

MoFO: Momentum-Filtered Optimizer for Mitigating Forgetting in LLM Fine-Tuning



Yupeng Chen* Senmiao Wang* Yushun Zhang Zhihang Lin Haozhe Zhang Weijian Sun Tian Ding Ruoyu Sun



香港中文大學 (深圳)
The Chinese University of Hong Kong, Shenzhen

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深圳市大数据研究院
Shenzhen Research Institute of Big Data

Background of Forgetting

Problem: LLMs lose pre-trained knowledge during fine-tuning.

Example: Fine-tuning Llama-2-7B on MetaMathQA

Downstream Benchmark: 14% → 50% (+36%)

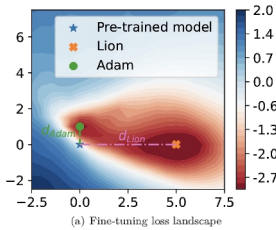
Pre-training Benchmark: 44% → 38% (-6%)

SFT Loss Landscape Finding

SFT landscape Finding:

LLM SFT loss landscape has distinct minima.

Different optimizers converge to distinct minima.

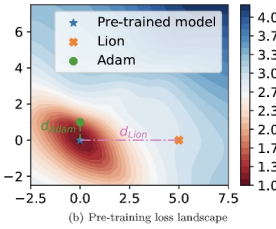


(a) Fine-tuning loss landscape

PT Landscape Finding:

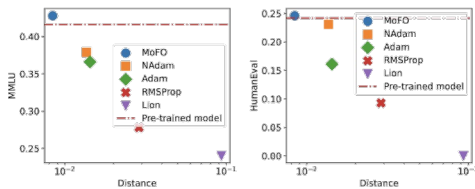
Minima Distance Matter:

$\|\theta_{\text{minima}} - \theta_{\text{pretrained}}\|$ is strongly correlated with the severity of forgetting.



(b) Pre-training loss landscape

Strong correlation between distance and forgetting (across different optimizers) in Llama-2-7B



Motivation for Algorithm Design

Design goals (multi-goal):

- (G1) Preservation of pretrained knowledge
- (G2) Strong fine-tuning performance

Key insight: Closer to pretrained params = less forgetting

Solution: Greedy Block Coordinate Descent (BCD)

MoFO Algorithm

Standard Adam:

$$\begin{aligned} t &\leftarrow t + 1, \quad g_t = \nabla_{\theta} f(\theta_{t-1}) \\ m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t \\ v_t &= \beta_2 v_{t-1} + (1 - \beta_2) (g_t \odot g_t) \\ \hat{m}_t &= \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t} \\ \theta_t &= \theta_{t-1} - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \end{aligned}$$

MoFO Update Rule:

$$[\text{FLT}_{\alpha}(m_t)]_i = \mathbf{1}(|(m_t)_i| \in \text{Top-}\alpha(|m_t|))$$

$$\theta_t = \theta_{t-1} - \eta_t \frac{\hat{m}_t \odot \text{FLT}_{\alpha}(m_t)}{\sqrt{\hat{v}_t}}$$

At each iteration, MoFO selects and updates only the most "influential" parameters (largest momentum magnitude), keeping others fixed.

Convergence Theory

Theorem 1 (Convergence of MoFO). Suppose that the first- and second-order momentum hyperparameters β_1 and β_2 satisfy $0 < \beta_1 < \sqrt{\beta_2} < 1$. The learning rate schedule at step t is $\eta_t = \eta/\sqrt{t}$ for some $\eta > 0$. Then, under Assumption 1, MoFO satisfies

$$\min_{0 \leq t \leq T-1} \|\nabla \mathcal{L}(\theta_t) \odot \text{FLT}_{\alpha}(\nabla \mathcal{L}(\theta_t))\|_1 = \mathcal{O}\left(\frac{\log T}{\sqrt{T}}\right) \quad \text{as } T \rightarrow \infty.$$

Moreover, this bound directly implies

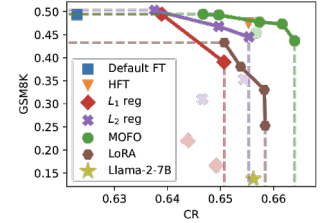
$$\min_{0 \leq t \leq T-1} \|\nabla \mathcal{L}(\theta_t)\|_p = \mathcal{O}\left(\frac{\log T}{\sqrt{T}}\right) \quad \text{as } T \rightarrow \infty,$$

for any $p \in [1, \infty]$.

MoFO achieves the same convergence rate as standard Adam for non-convex optimization.

Results: Better Pareto Front

MoFO consistently achieves a better Pareto front compared to all baseline methods.



All BCD methods alleviate forgetting.

MoFO achieves the best performance.

Method	GSMSK	General Capability			
		CR	MMLU	HEval	Avg.
Llama-2-7B	13.7	65.6	42.0	24.2	43.9
Default FT	49.4	62.3	36.6	16.1	38.3
Random BCD	35.0	65.8	41.1	25.1	44.0
Grad BCD	40.2	66.0	41.6	28.0	45.2
MV BCD	42.2	66.0	40.0	27.6	44.5
MoFO	45.4	65.7	43.5	27.4	45.5

Continual Learning and RLHF

Sequential SFT Continual Learning

	OP	BWT
Default FT	38.4	-10.3
HFT	39.9	-10.1
MoFO	41.3	-5.4
GEM	40.8	-8.5
GEM + MoFO	41.7	-6.7
EWC	41.1	-8.3
EWC + MoFO	43.2	-4.4
Replay	45.5	4.7
Replay + MoFO	47.0	4.8

RLHF (DPO)

	MMLU	GPQA
Pretrained LLM	72.9	21.7
Default FT	71.8	24.8
MoFO	73.4	25.8

MoFO is **orthogonal** to existing CL methods.

Try MoFO



Paper